

AP CSP | Chapter 8 Practice

Diagnostic | 10 questions | Suggested time: 18 minutes

All questions are original classroom practice. These short sets are not a full AP exam. Work without a calculator. The symbol ← means assignment; list indices start at 1. Select ONE answer unless a question says Select TWO. For this practice, a two-answer item counts correct only when both choices, and no others, are selected.

Name / private identifier: _____ Date: _____

Q01 | A fictional community-fridge app records food donations and shows nearby pickup locations. Which statement describes the purpose of the app rather than a specific function?

- A. Display pickup locations in distance order.
- B. Help residents share surplus food with people who can use it.
- C. Read a donor's selected food category.
- D. Store the time when a donation was entered.

My selection: _____

Q02 | A team's fictional booking app passed its own tests, but several intended users cannot complete a booking. Which TWO actions best support an iterative response to this evidence? Select TWO.

- A. Observe intended users attempting a booking and ask where they encounter difficulty.
- B. Add more features before investigating the failed bookings.
- C. Revise the design based on observed barriers, then test the revised booking task.
- D. Keep the design because the developers already completed the task successfully.

My selection: _____

Q03 | A sensor system needs a different fixed-length binary ID for each of 19 devices. What is the smallest number of bits per ID that provides enough distinct patterns?

- A. 4 bits
- B. 19 bits
- C. 6 bits
- D. 5 bits

My selection: _____

Q04 | A laboratory must be able to reconstruct every original recorded sensor value exactly after archiving the data. Which approach satisfies this requirement?

- A. Use a lossless method that reduces size while preserving exact reconstruction.
- B. Use a lossy method whenever its graph looks unchanged.
- C. Round every value to the nearest whole number before compression.
- D. Keep only the average of each group of ten readings.

My selection: _____

Q05 | What value is displayed by this code?

```
a ← 8  
b ← a  
a ← a + 3  
DISPLAY(a + b)
```

- A. 16
- B. 22
- C. 19
- D. 11

My selection: _____

Q06 | A program records an unknown number of daily temperatures. Why does storing them in a list help manage the program's complexity?

- A. It guarantees that every recorded temperature is accurate.
- B. One traversal can process a changing number of values without adding a separately named variable for each day.
- C. It makes every search take exactly one operation.
- D. It removes the need to decide how missing readings should be handled.

My selection: _____

Q07 | A network has links A-B, B-D, A-C and C-D. After link A-B fails, which statement best explains why A can still send data to D?

- A. The network automatically encrypts the failed link.
- B. Every packet must still travel through B.
- C. D can receive data without any remaining connection to A.
- D. The remaining route A-C-D provides redundancy.

My selection: _____

Q08 | Three independent tasks take 2, 6 and 7 seconds on one processor. Two identical processors can each run one task at a time; tasks cannot be split, and there is no overhead. What is the shortest time to finish all three?

- A. 7 seconds
- B. 8 seconds
- C. 7.5 seconds
- D. 15 seconds

My selection: _____

Q09 | A fictional town makes a service available only through a smartphone app. It then adds staffed public kiosks where residents can use the service without a personal device or internet subscription. Which conclusion is best supported?

- A. The kiosks can reduce device and connection barriers, but opening hours and travel may still limit access.
- B. The kiosks eliminate every cause of the digital divide.
- C. Every resident will now choose to use the service.
- D. A resident without a smartphone must still buy one to use the kiosk.

My selection: _____

Q10 | A fictional speech-recognition system was trained mostly on recordings from one accent group and makes more errors for another group. Which response most directly investigates and reduces this problem?

- A. Add many more recordings from only the already well-represented group.
- B. Stop measuring errors separately for different accent groups.
- C. Collect relevant recordings from underrepresented speakers and evaluate error rates across groups after revision.
- D. Remove accent-group labels from the evaluation and assume that bias is gone.

My selection: _____

AP CSP | Chapter 8 Practice

Algorithms | 10 questions | Suggested time: 20 minutes

All questions are original classroom practice. These short sets are not a full AP exam. Work without a calculator. The symbol $\leftarrow$ means assignment; list indices start at 1. Select ONE answer unless a question says Select TWO. For this practice, a two-answer item counts correct only when both choices, and no others, are selected.

Name / private identifier: _____ Date: _____

Q11 | What value is assigned to result?

```
result  $\leftarrow$  (23 MOD 6) + 15 / 4
```

- A. 8
- B. 8.75
- C. 6.75
- D. 5

My selection: _____

Q12 | The code is intended to exchange the original values of left and right. Which statement should replace MISSING?

```
left  $\leftarrow$  4  
right  $\leftarrow$  9  
saved  $\leftarrow$  left  
left  $\leftarrow$  right  
MISSING
```

- A. left $\leftarrow$ saved
- B. right $\leftarrow$ left
- C. saved $\leftarrow$ right
- D. right $\leftarrow$ saved

My selection: _____

Q13 | For numeric x, which expression is true exactly when x is outside the inclusive range from 10 to 20?

- A. NOT ((x $\geq$ 10) AND (x $\leq$ 20))
- B. (x < 10) AND (x > 20)
- C. (x $\leq$ 10) OR (x $\geq$ 20)
- D. NOT ((x < 10) OR (x > 20))

My selection: _____

Q14 | What value is displayed after the loop?

```
total ← 0
value ← 2
REPEAT 4 TIMES
{
  total ← total + value
  value ← value + 2
}
DISPLAY(total)
```

- A. 8
- B. 12
- C. 20
- D. 30

My selection: _____

Q15 | What value is displayed after the loop terminates?

```
x ← 1
steps ← 0
REPEAT UNTIL (x ≥ 10)
{
  x ← x * 2
  steps ← steps + 1
}
DISPLAY(steps)
```

- A. 3
- B. 4
- C. 5
- D. 10

My selection: _____

Q16 | What is the final value of values? List indices start at 1.

```
values ← [4, 8, 12]
INSERT(values, 2, 6)
REMOVE(values, 3)
APPEND(values, values[1])
```

- A. [4, 6, 8, 4]
- B. [4, 8, 12, 4]
- C. [6, 8, 12, 6]
- D. [4, 6, 12, 4]

My selection: _____

Q17 | What value is displayed by the calls to combine?

```
PROCEDURE combine(a, b)
{
  RETURN(2 * a + b)
}
x ← combine(3, 5)
DISPLAY(combine(x, 1))
```

- A. 23
- B. 12
- C. 11
- D. 13

My selection: _____

Q18 | Binary search looks for 21 in the sorted list [3, 8, 12, 15, 21, 28, 34]. It checks the middle item of the remaining range, discards the checked item when it does not match, and stops at a match. Which sequence of item values is checked?

- A. 3, 8, 12, 15, 21
- B. 15, 21
- C. 15, 28, 21
- D. 34, 28, 21

My selection: _____

Q19 | For input size n , algorithm P takes $3n + 2$ steps and algorithm Q takes n^2 steps. Both solve the same problem correctly. Which statement follows from the AP CSP growth-rate notion of reasonable time?

- A. Only P is reasonable because Q contains an exponent.
- B. Both have polynomial step counts and are considered reasonable, although Q grows faster.
- C. Q is exponential because its formula is n^2 .
- D. Both must finish within one second for every input size.

My selection: _____

Q20 | A robot starts at row 3, column 2, facing north on a 4-by-4 grid. Rows increase southward and columns increase eastward. Only cell (2, 2) is blocked. What is the final position and direction after this code?

```
IF (CAN_MOVE(forward))  
{  
    MOVE_FORWARD()  
}  
ELSE  
{  
    ROTATE_RIGHT()  
    MOVE_FORWARD()  
}  
ROTATE_LEFT()  
MOVE_FORWARD()
```

- A. Row 3, column 3, facing east
- B. Row 2, column 2, facing north
- C. Row 2, column 3, facing east
- D. Row 2, column 3, facing north

My selection: _____

AP CSP | Chapter 8 Practice

Mixed practice | 20 questions | Suggested time: 34 minutes

All questions are original classroom practice. These short sets are not a full AP exam. Work without a calculator. The symbol ← means assignment; list indices start at 1. Select ONE answer unless a question says Select TWO. For this practice, a two-answer item counts correct only when both choices, and no others, are selected.

Name / private identifier: _____ Date: _____

Q21 | Fictional scenario: RepairLink aims to help residents arrange successful repairs with volunteer fixers. Residents enter an item category and available times; the app suggests volunteers. Its first user study included only people using desktop computers on weekdays. Which additional evidence would most directly help evaluate whether the app is fulfilling its stated purpose?

- A. The number of lines in the app's source code.
- B. The number of different colors used in the interface.
- C. The proportion of repair requests that result in an agreed appointment and a completed repair.
- D. The alphabetical order of the volunteers' names.

My selection: _____

Q22 | Fictional scenario: RepairLink aims to help residents arrange successful repairs with volunteer fixers. Residents enter an item category and available times; the app suggests volunteers. Its first user study included only people using desktop computers on weekdays. Which TWO actions would best broaden evidence for the next design iteration? Select TWO.

- A. Assume the first study represents every intended user because it used real people.
- B. Observe intended users completing repair requests on smartphones and at other times.
- C. Collect more responses only from the same weekday desktop users, then claim all access barriers are resolved.
- D. Include intended users with relevant accessibility needs, identify task barriers, and retest revisions.

My selection: _____

Q23 | An uncompressed image has 8 columns and 6 rows of pixels. Each pixel uses 3 bits. Ignoring all headers, metadata and padding, how many bits are needed for the pixel data?

- A. 18 bits
- B. 48 bits
- C. 384 bits
- D. 144 bits

My selection: _____

Q24 | A 240 KB dataset is compressed by method A to 60 KB with some information discarded, or by method B to 90 KB with exact reconstruction possible. An archive requires every original value. Which choice meets the requirement, and why?

- A. Choose A because the smallest file must preserve the most information.
- B. Choose B because it preserves exact reconstruction even though its compressed file is larger.
- C. Choose either because every compression method is reversible.
- D. Reject B because a compressed file can never be restored exactly.

My selection: _____

Q25 | Five synthetic travel times are 12, 15, 18, 20 and 95 minutes. The 95-minute record has not been verified as an error. Which TWO statements are supported? Select TWO.

- A. The mean of all five values is 32 minutes.
- B. The median of all five values is 20 minutes.
- C. Removing 95 solely because it raises the mean proves that the remaining sample is unbiased.
- D. An analyst can report results with and without 95 as a labeled sensitivity check while investigating the record.

My selection: _____

Q26 | Two sensors record the same temperature as 20 and 68. Their metadata identifies the units as Celsius and Fahrenheit respectively. Before computing a combined mean, what should an analyst do?

- A. Discard 68 because its numerical value is larger.
- B. Average the raw values to obtain 44 and omit the units.
- C. Convert the values to a common temperature unit using the metadata.
- D. Convert both records to binary so that their units no longer matter.

My selection: _____

Q27 | With score initially equal to 80, what is displayed, in order?

```
IF (score ≥ 70)
{
  IF (score ≥ 90)
  {
    DISPLAY("Gold")
  }
  ELSE
  {
    DISPLAY("Silver")
  }
}
IF (score ≥ 70)
{
  DISPLAY("Ready")
}
```

- A. Silver, then Ready
- B. Gold, then Ready
- C. Silver only
- D. Ready only

My selection: _____

Q28 | What is the final value of totals?

```
readings ← [4, 2, 5]
totals ← []
total ← 0
FOR EACH reading IN readings
{
  total ← total + reading
  APPEND(totals, total)
}
```

- A. [4, 2, 5]
- B. [11, 11, 11]
- C. [0, 4, 6]
- D. [4, 6, 11]

My selection: _____

Q29 | Consider hasAbove. Which TWO statements about its returned values are correct? Select TWO.

```
PROCEDURE hasAbove(values, limit)
{
  FOR EACH value IN values
  {
    IF (value > limit)
    {
      RETURN(true)
    }
  }
  RETURN(false)
}
```

- A. hasAbove([9, 10], 9) returns false.
- B. hasAbove([4, 9], 9) returns false.
- C. hasAbove([], 9) terminates with an invalid-index error.
- D. hasAbove([9, 10], 9) returns true.

My selection: _____

Q30 | A simulation independently generates RANDOM(1, 8) once per trial and records a success when the result is at most 3. Which statement is most accurate for 10,000 trials?

- A. Exactly 3,750 successes are guaranteed.
- B. About 3,750 successes are expected, but the observed count can differ.
- C. Each trial has probability 3/7 of success because there are seven gaps between 1 and 8.
- D. Exactly three successes must occur in every block of eight trials.

My selection: _____

Q31 | The code should set found to true exactly when target appears anywhere in items. It can fail when a matching item is followed by a nonmatching item. Which change fixes this for every list, including an empty list?

```
found ← false
FOR EACH item IN items
{
  IF (item = target)
  {
    found ← true
  }
  ELSE
  {
    found ← false
  }
}
```

- A. Set found to true before the loop instead.
- B. Traverse the list in reverse order without changing the loop body.
- C. Remove the ELSE block, keeping the initial found ← false.
- D. Replace item = target with item > target.

My selection: _____

Q32 | Researchers prove that no algorithm can always give the correct yes-or-no answer to a certain question for every possible program and input. Which conclusion follows?

- A. The general problem is undecidable, although some individual cases may still be solved.
- B. The problem only needs a faster processor to become decidable.
- C. No individual case of the problem can ever be answered.
- D. A solution must exist because all answers are either yes or no.

My selection: _____

Q33 | Under a stated comparison-count model, method A uses 100 comparisons per search. Method B first sorts once using 500 comparisons, then uses 7 comparisons per search. For 10 searches of the unchanged dataset, which method uses fewer comparisons?

- A. A, by 430 comparisons.
- B. B, by 930 comparisons.
- C. A, because B must sort again before every search.
- D. B, by 430 comparisons.

My selection: _____

Q34 | A file is sent using TCP over the Internet. Packets arrive out of order and one packet is missing. Assuming communication can continue, which mechanism supports complete, correctly ordered delivery?

- A. Require every packet to have followed exactly the same route.
- B. Use sequence information to order the data and acknowledgments/retransmission to recover missing data.
- C. Replace the missing packet with a random packet of the same length.
- D. Use DNS to infer the contents of the missing packet.

My selection: _____

Q35 | A diagnostic request to a server's known IP address succeeds, but a DNS lookup for its domain name fails. Which function is most directly failing in this observation?

- A. Encrypting all traffic between the client and server.
- B. Storing every web page in the Internet's routers.
- C. Resolving the domain name to an IP address.
- D. Creating a physical connection between every pair of computers.

My selection: _____

Q36 | A synthetic training-app model has two groups: Group A is 40% of the intended population and Group B is 60%. Positive-rating rates are fixed at 90% for A and 30% for B. A constructed sample contains 50 people from each group. Which TWO statements follow from this model? Select TWO.

- A. The constructed sample's overall positive-rating rate is 60%.
- B. The intended population's rate must be 60% because the sample has equal group sizes.
- C. Applying the stated population proportions to the fixed group rates gives 54%.
- D. Reweighting the two groups guarantees that all self-selection and measurement bias has been removed.

My selection: _____

Q37 | Mira wants to send Lee a confidential message using public-key encryption. She has obtained an authentic copy of Lee's public key. Which pair of actions is appropriate?

- A. Mira encrypts with Lee's public key; Lee decrypts with Lee's private key.
- B. Mira encrypts with Lee's private key; every reader decrypts with Lee's public key.
- C. Mira encrypts with Mira's public key; Lee decrypts with Lee's unrelated private key.
- D. Mira publishes Lee's private key so that Lee can verify it.

My selection: _____

Q38 | A fictional citizen-science project invites volunteers to submit wildlife observations over the Internet. Most early reports come from a few easily reached parks. Which response best improves the usefulness of the growing dataset?

- A. Assume that a large total number of submissions guarantees geographic representativeness.
- B. Reject all location metadata to make the reports easier to compare.
- C. Treat every volunteer report as correct without verification.
- D. Validate submissions and recruit observers from less-covered areas while documenting remaining coverage limits.

My selection: _____

Q39 | An icon's stated license permits sharing and adaptation only for noncommercial uses and requires attribution. A club plans to sell an app containing a recolored version of the icon. Which action would meet the stated permissions for the planned use?

- A. Use the recolored icon without attribution because its colors changed.
- B. Choose a different icon whose license permits this commercial use, and satisfy that license's conditions.
- C. Use the original icon in the sold app because attribution alone cancels the noncommercial restriction.
- D. Use the icon because anything downloadable from the Web is public domain.

My selection: _____

Q40 | A fictional fitness app publishes individual route histories after removing users' names. Many routes repeatedly start and end at the same private residences. Which privacy concern remains?

- A. Removing names guarantees that no published information can be linked to a person.
- B. Converting the coordinates to a different file format guarantees anonymity.
- C. Route patterns may be combined with other information to infer a person's home or identity.
- D. The only possible risk is that the published files occupy too much storage.

My selection: _____